

PRIOR CASE RETRIEVAL FOR INDIAN LEGAL DOCUMENTS

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MASTER OF COMPUTER APPLICATIONS

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “Prior Case Retrieval for Indian Legal Documents” is a project work successfully done by Aniket Mani (205124013) and Rishu Kumar (205124078) in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications from the National Institute of Technology, Tiruchirappalli, during the academic year 2025-26.

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ABSTRACT

Prior-case retrieval identifies earlier judgments that are legally relevant to a new case. The task is difficult because Indian judgments are long, structurally heterogeneous, and relevant for reasons that extend beyond surface word overlap. This internship investigated Indian legal prior-case retrieval using lexical retrieval, legal entailment reranking, and rhetorical-role importance. The work first established BM25, TF-IDF, dense embedding, and role-aware baselines on the Indian Legal Text Understanding and Reasoning benchmark. Lexical retrieval remained a strong and reliable candidate-generation method, while dense single-vector models were limited by truncation and long-document evidence loss.

The second phase modelled relevance as legal entailment between the Facts of a query judgment and the Ratio of the Decision of a candidate judgment. BM25 was used to generate candidate cases, and weak supervision was used to create labelled query-candidate pairs for training an InLegalBERT reranker. Hybrid scoring showed that sparse retrieval and semantic entailment signals are complementary. The third phase studied which rhetorical-role interactions are most useful for distinguishing genuine legal support from misleading similarity. Role pairs involving precedent, ratio, facts, and arguments showed the strongest discrimination, while common statutory overlap was less informative. Overall, the project shows that effective Indian legal prior-case retrieval requires lexical evidence, legal entailment, and role-aware reasoning.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
BM25	Best Matching 25
CLS	Classification token representation
IL-PCR	Indian Legal Prior-Case Retrieval
IL-TUR	Indian Legal Text Understanding and Reasoning
MAP	Mean Average Precision
MRR	Mean Reciprocal Rank
nDCG	Normalized Discounted Cumulative Gain
PCR	Prior-Case Retrieval
RR	Rhetorical Role
SBERT	Sentence-BERT
TF-IDF	Term Frequency-Inverse Document Frequency

CHAPTER 1 - INTRODUCTION

This project investigates how information-retrieval and language-model techniques can be combined to retrieve legally relevant prior cases. A query is itself a judgment or case description, and the system must rank earlier candidate judgments that can support legal research. The central difficulty is that relevance depends on both surface wording and legal function.

We organized the internship into three connected phases. Phase 1 establishes lexical and role-aware baselines. Phase 2 introduces a Facts-to-Ratio entailment signal. Phase 3 studies rhetorical-role importance and a full-role-text ModernBERT reranker. Each phase produces evidence that informs the next.

Report principle: We distinguish reproduced results, executed notebook results and prepared but incomplete experiments.

1.1 Background

Legal professionals rarely search only for documents containing the same words. They search for earlier cases with comparable facts, legal questions, controlling principles, precedents or statutory interpretation. Conventional lexical retrieval is therefore useful but incomplete.

Indian judgments are particularly challenging for automated retrieval because they are long, contain multiple rhetorical sections and may repeat standard legal language. Two cases can cite the same statute without being relevant to each other, while two relevant cases may use different wording to express similar reasoning.

The IL-PCR task provides query cases, candidate prior cases and official relevance judgements. The IL-TUR annotations divide case text into roles such as Fact, Ratio of the Decision, Precedent, Argument, Statute and Ruling by the Present Court. Together, these resources support both document-level and role-aware experiments.

1.2 Problem Statement

The problem is to rank prior judgments for each query case so that officially relevant candidates appear as high as possible. A successful system must balance exact legal terminology with semantic and structural similarity.

A purely lexical system can overvalue shared citations, sections or boilerplate. A purely semantic system can lose exact statutory signals and may generalize according to pseudo-label definitions rather than official relevance. Long-document neural models also face substantial computational cost.

- Retrieve relevant prior cases from a large candidate collection.
- Use rhetorical roles without discarding full legal context.
- Evaluate with official qrels rather than relying only on generated labels.
- Document incomplete experiments without presenting them as final results.

1.3 Objectives

The internship objectives were translated into measurable retrieval, entailment and ranking experiments.

- Prepare and validate legal case text while preserving citations, sections and legally meaningful punctuation.
- Compare BM25, TF-IDF and prepared advanced baselines under a common evaluation framework.
- Measure the usefulness of Fact, Ratio, Precedent and cross-role similarity.
- Construct a Facts-to-Ratio entailment dataset and train an InLegalBERT cross-encoder.
- Compare lexical, neural and hybrid scores on official relevance.
- Analyse positive and negative Gemini role-importance datasets.
- Train a full-role-text cached ModernBERT ranking head and record its internal performance.
- Create reproducible result notebooks containing only tables and visualizations.

1.4 Scope and Contributions

The project covers offline experimentation on the supplied IL-PCR and IL-TUR resources. It does not deliver a production legal-search service, provide legal advice or claim that pseudo-label agreement is equivalent to official prior-case relevance.

Our main contributions are a reproduced BM25 and TF-IDF comparison, executed rhetorical-role embedding experiments, a Gemini-assisted entailment dataset, an official hybrid result, a positive-negative role analysis and a cached ModernBERT training pipeline.

Contribution	Evidence
Lexical baseline	TF-IDF exceeded BM25 under reproduced Phase 1 settings.
Role-aware retrieval	Learned role combination exceeded individual role pairs.
Entailment reranking	Hybrid BM25 and InLegalBERT produced best official Phase 2 F1.
Role weighting	Positive and hard-negative role patterns were quantified.
Long-text training	Cached full-role-text head improved internal validation MRR.

CHAPTER 2 - TECHNICAL BACKGROUND

This chapter summarizes the concepts used in the project: lexical information retrieval, ranking metrics, entailment, rhetorical roles and contextual encoders. The explanations focus on how each concept affected our design decisions rather than presenting a general survey.

The final pipeline is a cascade. A fast retriever proposes candidates, role-aware or entailment models provide additional evidence, and the candidates are ranked for official evaluation. This

separation is important because a computationally expensive model cannot feasibly score every query-candidate pair with full legal text.

2.1 Legal Prior-Case Retrieval

Prior-case retrieval is a ranking task. For each query case q and candidate case c , the system estimates a relevance score $s(q,c)$. Candidates are sorted in descending order, and the quality of the ordering is measured against official relevance judgements.

Unlike generic document search, legal relevance may depend on a combination of factual analogy, legal issue, ratio, authority and procedural context. A candidate can be topically similar yet legally irrelevant. This motivates the use of multiple signals.

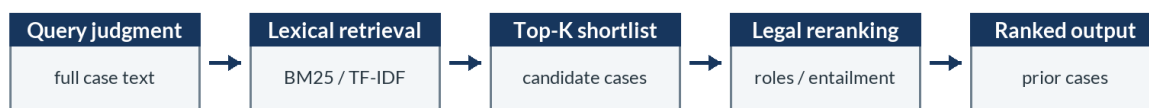


Figure 2.1: Retrieval cascade used in the project.

2.2 BM25 and TF-IDF

BM25 scores a candidate using query-term frequency, inverse document frequency, term-frequency saturation and document-length normalization. It is widely used as a first-stage retriever because it is efficient and rewards rare exact legal terminology.

TF-IDF represents each query and candidate as a weighted sparse vector. Cosine similarity normalizes vector length and measures the angle between them. In our reproduced setting, sublinear term frequency and cosine normalization were especially effective for long legal documents.

The two methods were evaluated as standalone full-text baselines. Later phases retained BM25 as an existing candidate-generation pipeline, but the report treats the stronger Phase 1 TF-IDF result as an important reason to compare or hybridize first-stage retrievers in future work.

2.3 Evaluation Metrics

We used several metrics because no single number completely describes a ranked list. Recall measures coverage, precision measures the concentration of relevant cases, and rank-sensitive metrics reward early retrieval.

Metric	Interpretation
Precision@K	Fraction of the top K candidates that are relevant.

Metric	Interpretation
Recall@K	Fraction of all relevant candidates found in the top K.
MRR	Reciprocal rank of the first relevant candidate, averaged by query.
MAP	Average precision across the positions of all relevant candidates.
nDCG@K	Discounted gain that rewards relevant cases near the top.
Micro-F1@K	Harmonic mean of micro precision and recall at K.
Hit@K	Whether at least one positive is found in a small validation group.

2.4 Legal Entailment

Phase 2 reframed one part of relevance as an entailment-like relation: do the facts stated in the query case receive support from the legal principle and reasoning in the candidate's Ratio of the Decision?

This is not formal logical entailment. It is a supervised relevance signal that asks whether the prior case's reasoning is applicable to the query facts. We used a cross-encoder so the two text segments could attend to each other jointly.

The formulation reduced some document-level noise but introduced two risks: truncation at 512 tokens and dependence on Gemini-generated pseudo-labels. Official evaluation was therefore essential.

2.5 Rhetorical Roles

Rhetorical-role annotation identifies the legal function of each sentence or passage. The main roles used in this project were Fact, Ratio of the Decision, Precedent, Argument, Statute and Ruling by the Present Court.

Same-role comparisons test whether corresponding legal functions align. Cross-role comparisons test relations such as query Fact to candidate Ratio or query Precedent to candidate Ratio. These comparisons produce interpretable features for retrieval and ranking.

Role	Function in a judgment
Fact	Events, parties and circumstances forming the dispute.
Ratio	Legal principle and reasoning necessary for the decision.
Precedent	Earlier authorities discussed or applied.
Argument	Claims advanced by the parties.
Statute	Legislation and sections relied upon.
Present Court Ruling	The current court's operative decision.

Role	Function in a judgment

CHAPTER 3 - DATASETS AND PREPARATION

We worked with two linked forms of the same legal collection. IL-PCR supplied full judgments and official prior-case links. The IL-TUR files used the same case identifiers but divided each judgment into rhetorical roles. Keeping the identifiers unchanged let us move from full-text retrieval to role-aware experiments without guessing which records belonged together.

Only the IL-PCR gold file was used to decide whether a retrieved case was officially relevant. Labels produced later with Gemini were kept in separate CSV files and were used only to prepare training pairs.

3.1 IL-PCR Dataset

In IL-PCR, a query and a candidate are ordinary UTF-8 text files. The file name is the case identifier. For the test split, one query is compared with a common pool of 1,727 candidate judgments, and the system returns those candidate identifiers in ranked order.

The local test archive contains 237 query files and 1,727 candidate files. The gold JSON has 237 query entries; 234 have at least one relevant case and three have an empty list. Across the non-empty entries there are 1,727 query-candidate relevance links.

Split	Queries	Candidates
Train	827	4,320
Development	118	1,023
Test	237	1,727

Identifier handling: Case identifiers were normalized consistently so qrels, text files and role files could be joined without self-retrieval.

3.2 IL-TUR Rhetorical-Role Data

The rhetorical-role copy is stored as JSON. Each file contains a case identifier followed by [passage, role] pairs. During loading, passages with the same label were joined so that, for example, the Facts of a query could be compared with the Ratio of a candidate.

Coverage is uneven. Fact, Ratio and Precedent appear in most candidate files, while lower-court rulings are less frequent. Missing-role handling is therefore required for role-pair scoring.

```
def load_case_file(path):
    data = json.load(open(path, encoding="utf-8"))
    case_id, rows = next(iter(data.items()))
    role_texts = defaultdict(list)
```

```

for row in rows:
    if len(row) >= 2:
        role_texts[row[1]].append(row[0])
return case_id, {
    role: " ".join(parts)
    for role, parts in role_texts.items()
}

```

Code 3.1: Representative rhetorical-role loader used in the notebooks.

3.3 One Retrieval Record, End to End

To show the data layout, we traced test query 0 001 104 022 through the files used by the notebooks. The query judgment asks whether payments for technical know-how are capital or revenue expenditure. Its gold list contains six earlier judgments. One of them, 0 000 594 383, deals with a limited right to use technical knowledge.

Part of the test record	Value in the original files
Query file	IL-PCR/ik_test/query/0 001 104 022.txt
Candidate pool	All 1,727 files under IL-PCR/ik_test/candidate/
Gold prior cases	0 000 594 383, 0 000 888 361, 0 000 923 991, 0 001 671 648, 0 001 805 387 and 0 001 990 557
Required prediction	A descending list of candidate IDs stored under 0 001 104 022.txt

Table 3.2: Complete retrieval record for test query 0 001 104 022.

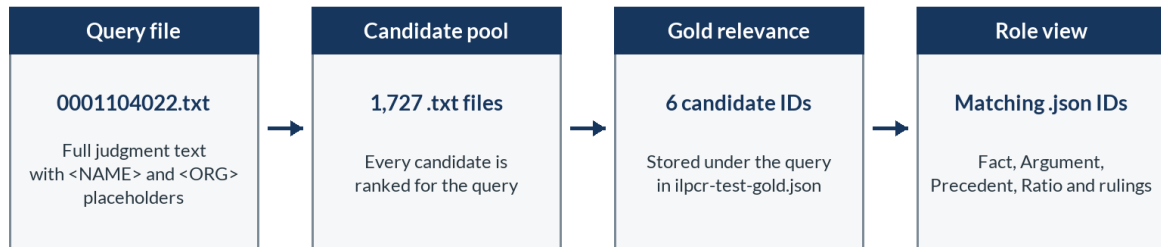


Figure 3.1: How the original text, gold list and role files are linked.

Original file	Verbatim excerpt (line breaks and spacing normalized)
Query 0001 104 022	“Whether ... the amount of Rs. 1,51,500 being the instalment payable ... under the agreement dated 14th July, 1965 is allowable as revenue expenditure?”
Relevant candidate 0000 594 383	“The assessee acquired ... merely the right to draw ... upon the technical knowledge ... for a limited period ... nor did the assessee acquire any asset or advantage of an enduring nature.”

Table 3.3: Excerpts from the original query and a gold candidate.

Case and role	Passage stored in the linked rhetorical-role JSON
Query 0001 104 022 - Fact	“These four references have been submitted ... in respect of the same assessee raising identical issues arising out of the same facts ...”
Query 0001 104 022 - Argument	“Expenditure for regularly operating the business cannot be considered to be the capital expenditure.”
Query 0001 104 022 - Ratio	“Mere right to use technical know-how without having proprietary right therein cannot be considered capital in the hands of the assessee.”
Candidate 0000 594 383 - Fact	“The assessee did not become entitled exclusively ... to the patents and trade marks; it acquired merely the right to draw ... upon the technical knowledge ... for a limited period.”
Candidate 0000 594 383 - Argument	“The assessee acquired under the agreement merely the right to draw ... upon the technical knowledge for a limited period ... nor did the assessee acquire any asset or advantage of an enduring nature.”

Table 3.4: The same query-candidate pair after rhetorical-role segmentation.

This is the actual transition used by the notebooks: the baseline indexed the complete .txt files, while the role experiments read passages from the matching .json files. The role labels were not equally available in every judgment, so a missing role remained empty and was excluded from that particular role-pair score.

Role	Query files	Candidate files	What the count means
Fact	231 / 237	1,687 / 1,727	A usable Fact passage was present
Argument	174 / 237	1,205 / 1,727	A party or counsel submission was present
Statute	121 / 237	905 / 1,727	A statutory passage was present
Precedent	223 / 237	1,538 / 1,727	A cited-case passage was present
Ratio	221 / 237	1,602 / 1,727	A reasoning passage was present

Role	Query files	Candidate files	What the count means
Lower-court ruling	59 / 237	382 / 1,727	The decision below was separately labelled
Present-court ruling	143 / 237	1,080 / 1,727	The final court ruling was separately labelled

Table 3.5: Judgment coverage for each rhetorical role.

3.4 Project Artifact Map

The table below summarizes the main artifacts used in this report. Generated ZIP files and model caches were treated as supporting artifacts, while notebooks and saved CSV/JSON summaries supplied the reported results.

Artifact	Phase	Purpose
IL-PCR/ik_train, ik_val, ik_test	1-3	Case text and official relevance.
pcr_roles and IL-TUR role files	1-3	Rhetorical-role text.
legal-case-retrieval-phase1-v2.ipynb	1	Expanded baseline methodology.
phase-2.ipynb	1/3	Role embeddings and learned weights.
legal-case-entailment*.ipynb	2	Dataset creation and cross-encoder.
phase2_results.json	2	Saved pseudo-label evaluation.
Phase 3 positive/negative CSVs	3	Role importance and hard negatives.
pcr_modernbert_l4_training_summary	3	Cached-head training record.
results/phase1-3.ipynb	All	Executed visualization notebooks.

3.5 Cleaning and Validation

Cleaning was conservative. We normalized repeated whitespace and extraction artifacts while preserving punctuation, section numbers, citations and legal terminology. These tokens can be decisive for retrieval.

Role validation removed missing placeholders and empty role sections before expensive labelling or embedding. Phase 2 loaded 827 queries, of which 799 had valid Fact text, and 4,319 candidates, of which 3,882 had valid Ratio text.

```
def clean_text(text, keep_punctuation=False):
    text = join_text(text)
    text = re.sub(r"\s+", " ", text).strip()
    text = re.sub(r"<[^\>]+\>", " ", text)
    if not keep_punctuation:
        text = re.sub(r"^[a-zA-Z0-9\s]*", " ", text)
    return re.sub(r"\s+", " ", text).strip().lower()
```

Code 3.3: Simplified cleaning logic from the Phase 1 notebook.

3.6 Qrels and Evaluation Protocol

Official qrels map each query identifier to one or more relevant candidate identifiers. During evaluation, self-matches were removed and the ranked list was truncated at each requested cutoff.

We report query-averaged metrics for retrieval quality and Micro-F1 for the Phase 2 official comparison. Internal Phase 3 training groups contain one positive and seven negatives; therefore MRR, Hit@1, Hit@3 and mean positive rank are more informative than full-corpus recall during training.

```
def recall_at_k(retrieved, relevant, k=10):
    if not relevant:
        return 0.0
    hits = len(set(retrieved[:k]) & set(relevant))
    return hits / len(relevant)

def precision_at_k(retrieved, relevant, k=10):
    return len(set(retrieved[:k]) & set(relevant)) / k
```

Code 3.4: Core top-K evaluation logic.

3.7 Result Provenance and Reproducibility

Not every saved notebook contains executed outputs. To avoid overstating progress, the report assigns each claim to one of three provenance categories.

Category	Meaning	Examples
Reproduced	Rerun from saved project data for this report.	BM25 versus TF-IDF baseline.
Executed	Output is present in a saved notebook or summary.	Role embeddings, Phase 2 training, ModernBERT internal metrics.
Prepared	Code or method exists but the saved output is absent or incomplete.	TextRank, InLegalBERT dense retrieval, official Phase 3 neural reranking.

Reproducibility: The results notebooks contain visualization code only and do not retrain or retest models.

CHAPTER 4 - PHASE 1: BASELINE RETRIEVAL

Phase 1 established the lexical baseline and investigated whether rhetorical-role structure can improve retrieval. The expanded Phase 1 notebook also prepared TextRank, SBERT, InLegalBERT and hybrid experiments; however, its saved output cells were not executed. We therefore report numerical values only from the reproduced baseline and executed role experiments.

The main result is that TF-IDF was stronger than BM25 under the reproduced full-text settings, while a learned combination of rhetorical-role similarities improved over individual role pairs.

4.1 Phase 1 Methodology

Candidate documents were cleaned, indexed and ranked for each query. The evaluation pipeline computed Precision, Recall, MRR, MAP and nDCG at multiple cutoffs.

The prepared experiment matrix included full-text BM25, full-text TF-IDF, TextRank plus TF-IDF, role-aware TF-IDF, role-aware BM25, SBERT, InLegalBERT, role-specific SBERT and lexical-semantic hybrids.

Method	Representation	Saved evidence
BM25	Full cleaned text	Reproduced metrics
TF-IDF	Sparse cosine vectors	Reproduced metrics
TextRank + TF-IDF	Extractive summary	Prepared code
Role-aware TF-IDF/BM25	Selected roles	Prepared code
MiniLM role embeddings	384-dimensional role vectors	Executed metrics
InLegalBERT dense retrieval	Legal contextual vectors	Prepared code
Weighted role score	Learned feature combination	Executed metrics

4.2 Dataset Scale

The collection is highly asymmetric: each split contains many more candidate judgments than query judgments. The test split used for the reproduced lexical comparison has 237 queries and 1,727 candidates.

This scale is manageable for sparse baselines but becomes expensive when every pair expands into several rhetorical roles and many neural windows.

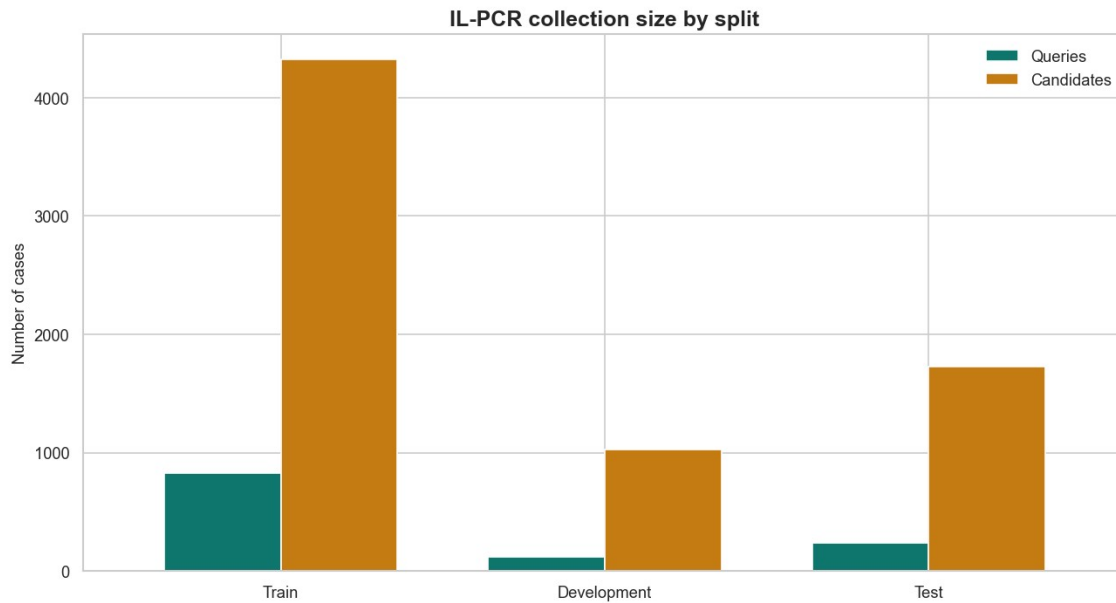


Figure 4.1: IL-PCR query and candidate counts by split.

4.3 BM25 Baseline

BM25 tokenizes each candidate and scores query terms using inverse document frequency, term-frequency saturation and document-length normalization. The method captures exact legal language, including rare citations and statutory phrases.

In the reproduced test, BM25 achieved MRR 0.4252, Recall@10 0.2194 and MAP 0.1492. It remained useful as a first-stage retriever in later pipelines, but it was not the strongest standalone Phase 1 method.

```
tokenized_corpus = [text.split() for text in candidate_texts]
bm25 = BM25Okapi(tokenized_corpus)

def bm25_retrieve(query_text, k=10):
    scores = bm25.get_scores(query_text.split())
    ranking = np.argsort(scores)[::-1]
    return [candidate_ids[i] for i in ranking[:k]]
```

Code 4.1: BM25 indexing and retrieval pattern.

4.4 TF-IDF Baseline

TF-IDF used unigram features, English stop-word filtering, sublinear term frequency and cosine similarity. The reproduced vocabulary was limited to 100,000 features.

Cosine normalization reduced the effect of long judgments, while sublinear term frequency reduced the impact of repeated boilerplate. TF-IDF achieved MRR 0.7385, Recall@10 0.5700 and MAP 0.4376.

```
vectorizer = TfidfVectorizer(
    stop_words="english",
    sublinear_tf=True,
    max_features=100000
)
candidate_matrix = vectorizer.fit_transform(candidate_texts)
```

```
query_vector = vectorizer.transform([query_text])
scores = cosine_similarity(query_vector, candidate_matrix)[0]
```

Code 4.2: Reproduced TF-IDF cosine retrieval configuration.

4.5 Common Evaluation Code

Both baselines were evaluated against the same candidate identifiers and official qrels. The common evaluator is important because differences in self-match removal, query cleaning or denominator handling can change the reported metrics.

The Phase 1 notebook also contains functions for MAP, MRR and nDCG. These utilities make it possible to compare future TextRank, dense and hybrid methods under the same protocol.

```
for query in queries:
    retrieved = retriever_fn(query, k=max(k_values))
    relevant = query["relevant_candidates"]
    for k in k_values:
        scores[k]["precision"].append(
            precision_at_k(retrieved, relevant, k)
        )
        scores[k]["recall"].append(
            recall_at_k(retrieved, relevant, k)
        )
```

Code 4.3: Shared evaluation loop.

4.6 BM25 versus TF-IDF at K=10

TF-IDF outperformed BM25 on every reported metric at K=10. The largest differences were in Recall, MRR and MAP. This indicates that the improvement was not limited to retrieving a single early relevant case.

The result is specific to the reproduced preprocessing and parameter choices. It does not prove that every BM25 configuration is inferior, but it makes TF-IDF a necessary candidate for future first-stage retrieval.

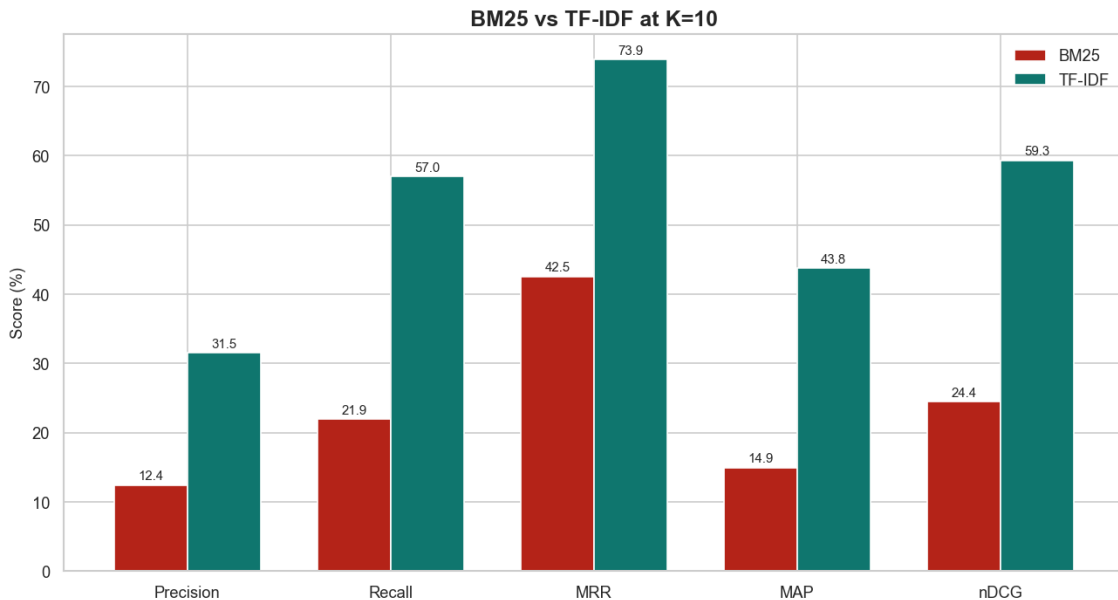


Figure 4.2: Reproduced BM25 and TF-IDF performance at K=10.

The executed Phase 1 notebook also recorded the following train-set results. The first two rows use the full IL-PCR training corpus (827 queries and 4,320 candidates). The remaining rows use the 100-document rhetorical-role subset with leave-one-out evaluation and silver relevance labels. They are reported together because they belong to the same notebook, but the two groups should not be compared as if they used the same test collection.

Method	Recall@10	Precision@10	MAP@10	MRR
BM25 (full text)	0.2369	0.1266	0.1456	0.3709
TF-IDF (full text)	0.4704	0.2398	0.3353	0.6051
Role-aware TF-IDF (symmetric)	0.8300	0.2490	0.5979	0.7947
Role-aware TF-IDF (Facts to Ratio)	0.7133	0.2140	0.4589	0.7000
BM25 (Facts only)	0.6733	0.2020	0.4493	0.6926

Table 4.2: Retrieval summary saved by the Phase 1 notebook.

4.7 Retrieval Depth Analysis

TF-IDF maintained its advantage as the cutoff increased. At K=50, TF-IDF Recall was 0.7375, compared with 0.3549 for BM25. TF-IDF MAP reached 0.4848, while BM25 MAP reached 0.1642.

A first-stage retriever should maximize the chance that relevant candidates reach the later reranker. These curves therefore suggest that a TF-IDF or BM25-TF-IDF hybrid shortlist should be evaluated before fixing the retriever for future experiments.

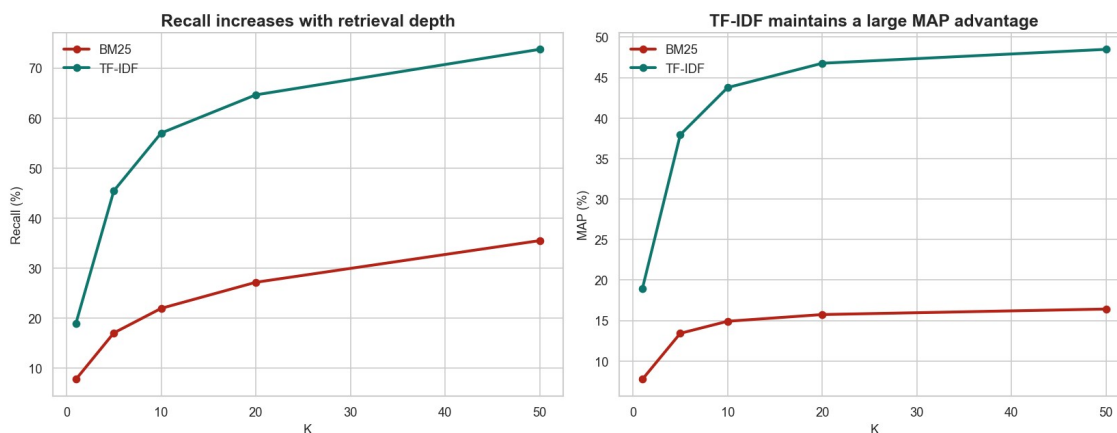


Figure 4.3: Recall and MAP across retrieval cutoffs.

4.8 Interpretation of the Lexical Result

The likely advantage of TF-IDF came from cosine length normalization, sublinear term frequency and the use of the complete document vocabulary. Long judgments contain repeated headings and boilerplate; saturating or normalizing these effects is essential.

BM25 was retained later primarily because the candidate-generation and labelling pipeline had already been implemented around it and because it is a conventional fast retriever for exact legal terminology. This is a workflow decision, not evidence that BM25 was the best Phase 1 method.

Methodological limitation: The later phases should be repeated with equal-size BM25, TF-IDF and hybrid shortlists.

4.9 Rhetorical-Role Coverage

Fact, Ratio and Precedent occur in most candidate role files. Argument, present-court ruling and Statute are less frequent, while lower-court rulings are comparatively sparse.

Role-aware systems must handle missing roles explicitly. In our experiments, a missing role produced no useful similarity rather than forcing an unrelated fallback passage.

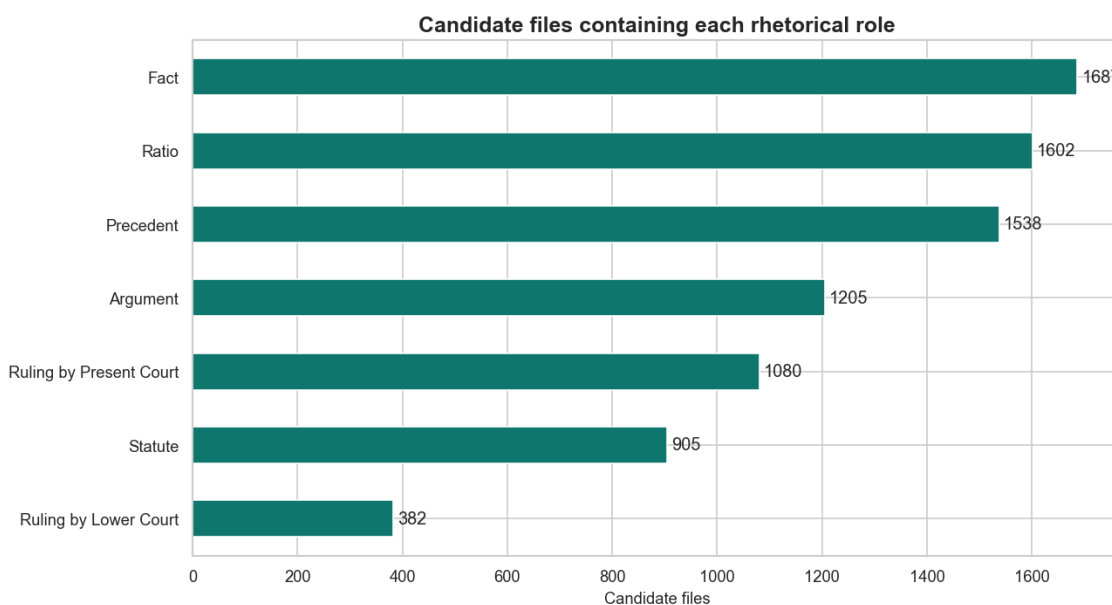


Figure 4.4: Candidate files containing each rhetorical role.

4.10 Single Role-Pair Retrieval

Precedent-to-Precedent was the strongest individual role pair by MRR, MAP and nDCG@10. Fact-to-Fact produced similar Recall@10, while Ratio-to-Ratio remained useful but weaker as an isolated signal.

Issue-to-Ratio performed very poorly, indicating that this role combination was either too sparse or too indirect for reliable retrieval in the available data.

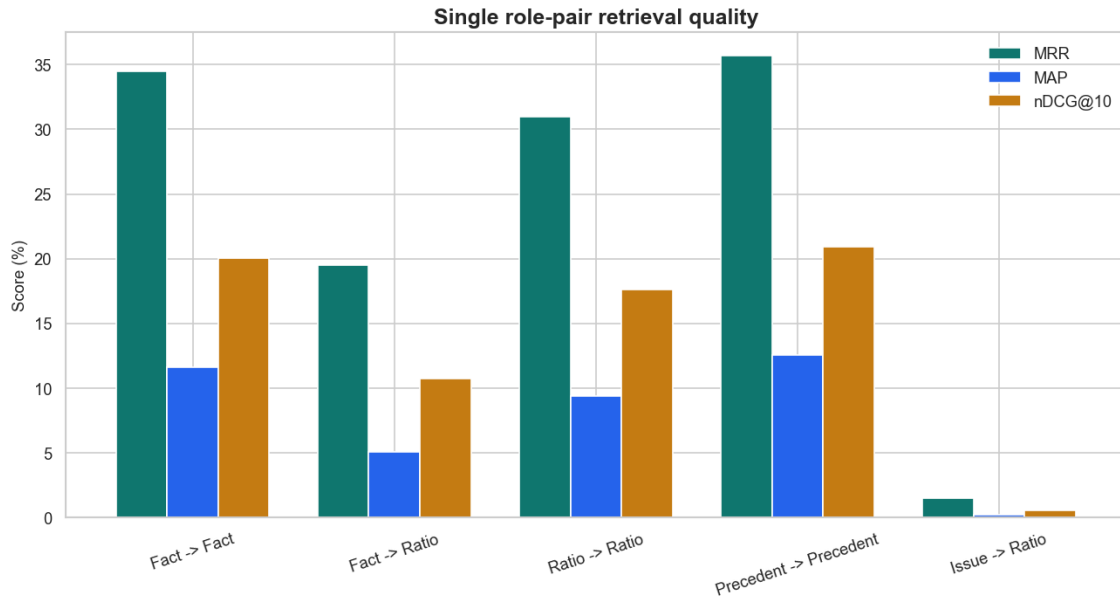


Figure 4.5: Retrieval quality for individual rhetorical-role pairs.

4.11 Learned Role Weights

We created 10,362 feature rows containing 1,727 positive and 8,635 sampled negative pairs. Positive pairs had higher average similarity on every principal role feature.

A query-group split produced ROC-AUC 0.7546. Fact-to-Fact received the largest learned weight of 5.802, followed by Ratio-to-Ratio at 2.457 and Precedent-to-Precedent at 1.795.

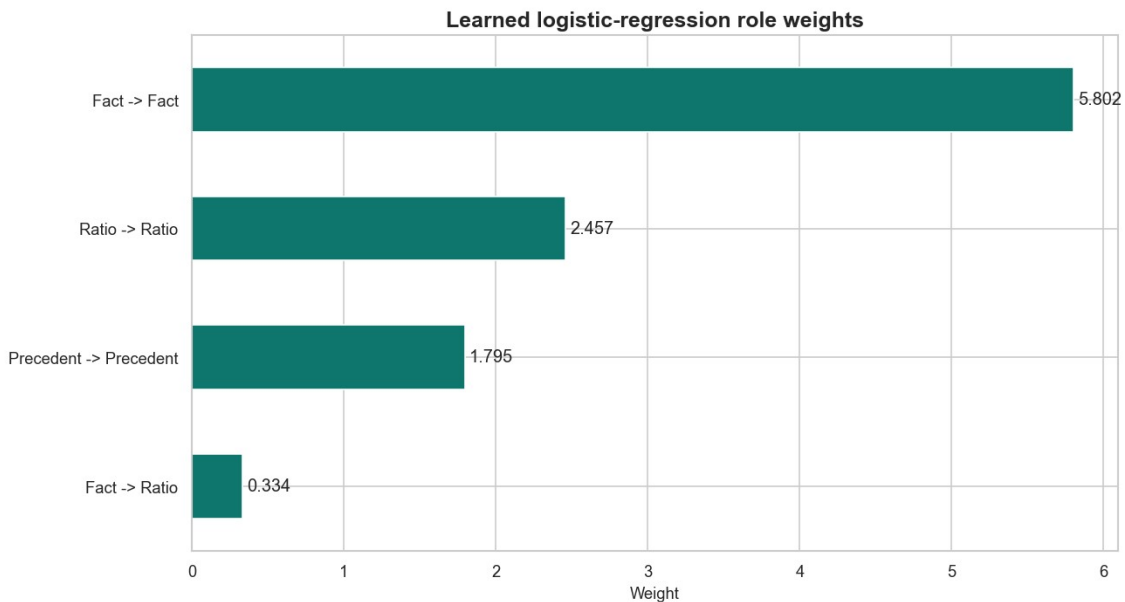


Figure 4.6: Learned role-similarity weights.

4.12 Phase 1 Summary

The learned weighted score improved Recall@10 from 0.2013 for the best single role to 0.2641, and improved MRR from 0.3572 to 0.4329. This confirms that factual, precedential and ratio evidence are complementary.

Phase 1 established two conclusions for the later work: lexical evidence remains strong, and legal structure should be combined rather than represented by one role alone.

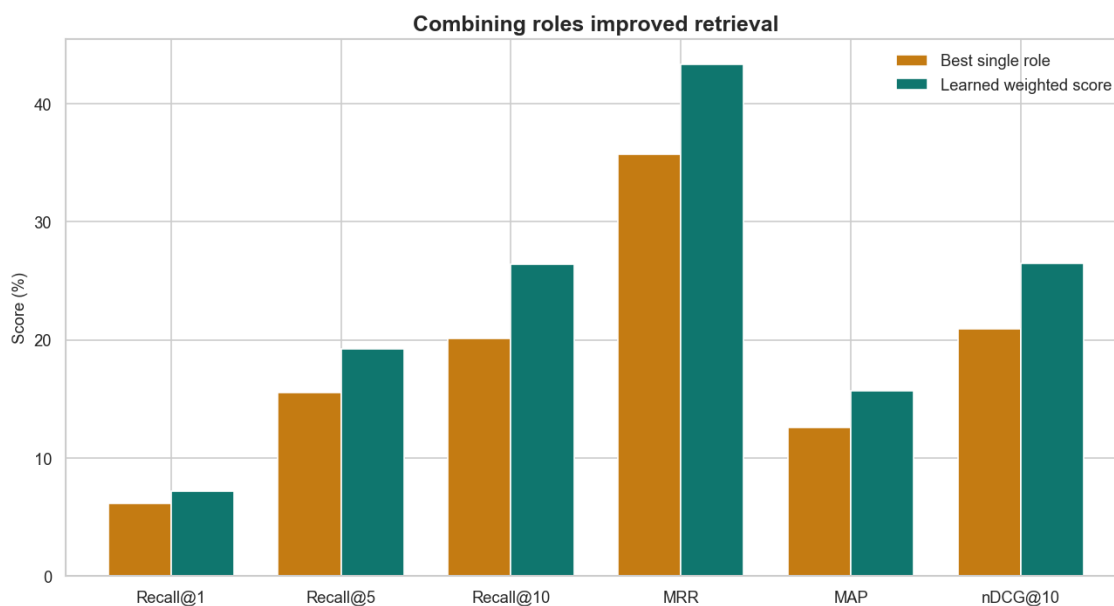


Figure 4.7: Best single role compared with the learned weighted score.

CHAPTER 5 - PHASE 2: LEGAL ENTAILMENT RERANKING

Phase 2 introduced an entailment-oriented score between query Facts and candidate Ratio of the Decision. The goal was to test whether legal reasoning in a prior case supports the circumstances described in the query case.

We created a BM25 candidate pool, generated Gemini pseudo-labels, trained an InLegalBERT cross-encoder and evaluated lexical, neural and hybrid rankings. The final official result favoured score fusion rather than the pure reranker.

5.1 Facts-to-Ratio Formulation

The query input is the Fact section: what happened, which parties were involved and what circumstances define the dispute. The candidate input is the Ratio of the Decision: the legal principle and reasoning used by the earlier court.

A positive label means the candidate reasoning provides meaningful legal support for the query facts. This formulation is narrower and more interpretable than comparing two entire judgments.

[QUERY FACTS]
<facts extracted from the query case>

[CANDIDATE RATIO]
<ratio of the decision from the prior case>

Label: support / non-support

Figure 5.1: Phase 2 pair formulation.

5.2 Dataset Construction Pipeline

The dataset was built in stages: validate role text, retrieve a top-50 BM25 pool, remove self-matches, request Gemini labels and save the cleaned result. This reduced the number of expensive model calls.

- 827 query role records were loaded; 799 contained valid Fact text.
- 4,319 candidate role records were loaded; 3,882 contained valid Ratio text.
- The initial top-50 pool contained 39,950 pairs.
- Removing 679 self-matches produced 39,271 candidate pairs.
- The final cleaned file contained 39,074 labels covering 795 queries.

5.3 Role Validation and Candidate Pool

Role validation prevented empty or malformed text from entering the labelling and training stages. The BM25 pool concentrated on lexically plausible candidates, making the negative examples harder than uniformly random pairs.

The pool is nevertheless influenced by BM25. Relevant cases not retrieved in the top 50 could never receive a positive pseudo-label, and 60 labelled queries had no positive candidate.

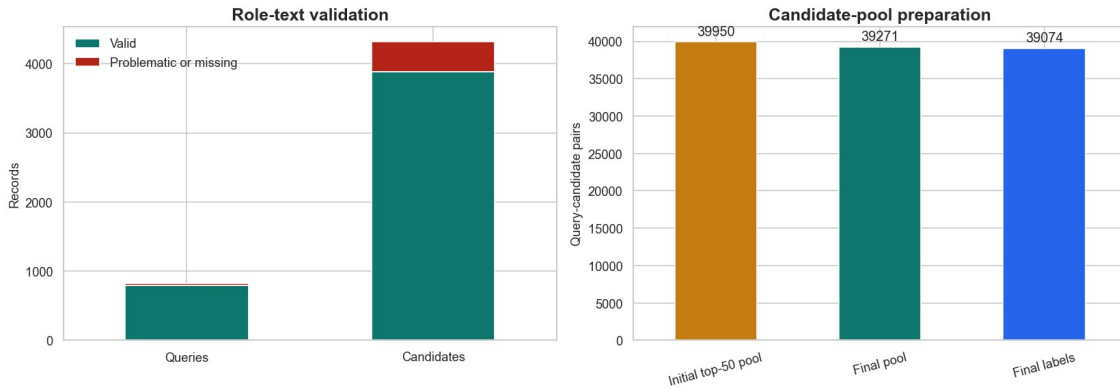


Figure 5.2: Role validation and candidate-pool preparation.

5.4 Gemini Pseudo-Labels

Gemini produced 3,384 positive and 35,690 negative labels. The positive rate was 8.66%, producing substantial class imbalance.

Generated labels made large-scale supervision possible, but they reflect the prompt's definition of support. They are not equivalent to the official prior-case relevance judgements.

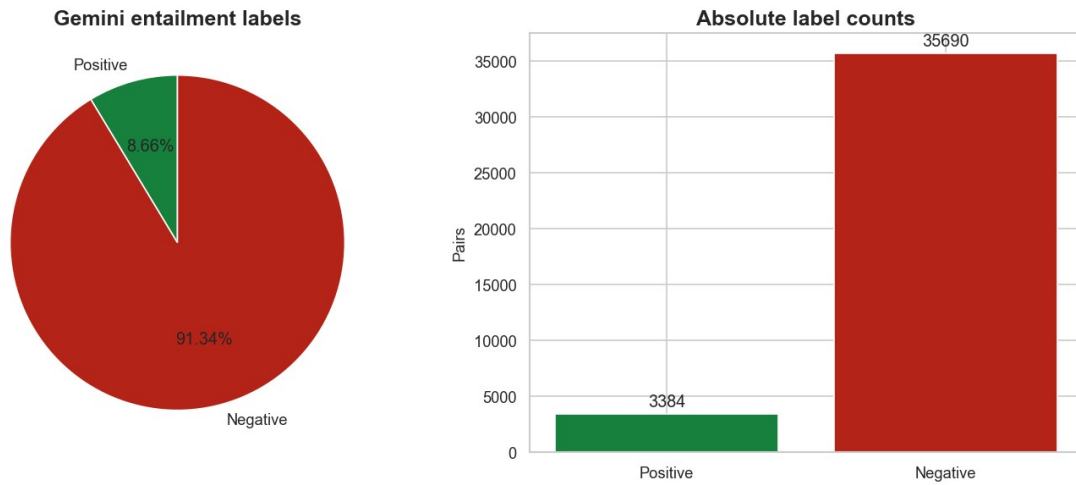


Figure 5.3: Distribution of Gemini entailment pseudo-labels.

5.5 Pair Creation Code

Role dictionaries were loaded from JSON, and Ratio text was selected for each candidate. A fallback to combined role text was used only during early dataset preparation when Ratio was missing.

```
candidate_ids = []
candidate_texts = []

for cid, roles in candidate_cases.items():
    ratio = roles.get("RatioOfTheDecision", "")
    if not ratio.strip():
        ratio = " ".join(roles.values())
    candidate_ids.append(cid)
    candidate_texts.append(ratio)
```

Code 5.1: Candidate Ratio extraction pattern.

Final filtering: The cleaned training dataset used only pairs that passed the role-text validation rules.

5.6 InLegalBERT Cross-Encoder

The smaller recorded training experiment used 4,635 pairs: 3,708 for training and 927 for validation. The positive rate was 14.67%. Query Facts and candidate Ratio were read jointly by a CrossEncoder based on law-ai/InLegalBERT.

The input was truncated to 512 tokens. This is a significant limitation because decisive facts or reasoning can occur later in a long legal section.

```
model = CrossEncoder(
    "law-ai/InLegalBERT",
    num_labels=1,
    max_length=512
)

model.fit(
    train_dataloader=train_dataloader,
    evaluator=evaluator,
    epochs=1,
    evaluation_steps=100
)
```

5.7 Training History

Validation F1 increased from 0.5639 at step 100 to 0.6586 at step 1,100. Average Precision increased from 0.5628 to 0.6578, although both curves fluctuated during training.

The result indicates that the model learned the pseudo-labelled support task. It does not yet establish improvement on official prior-case relevance.

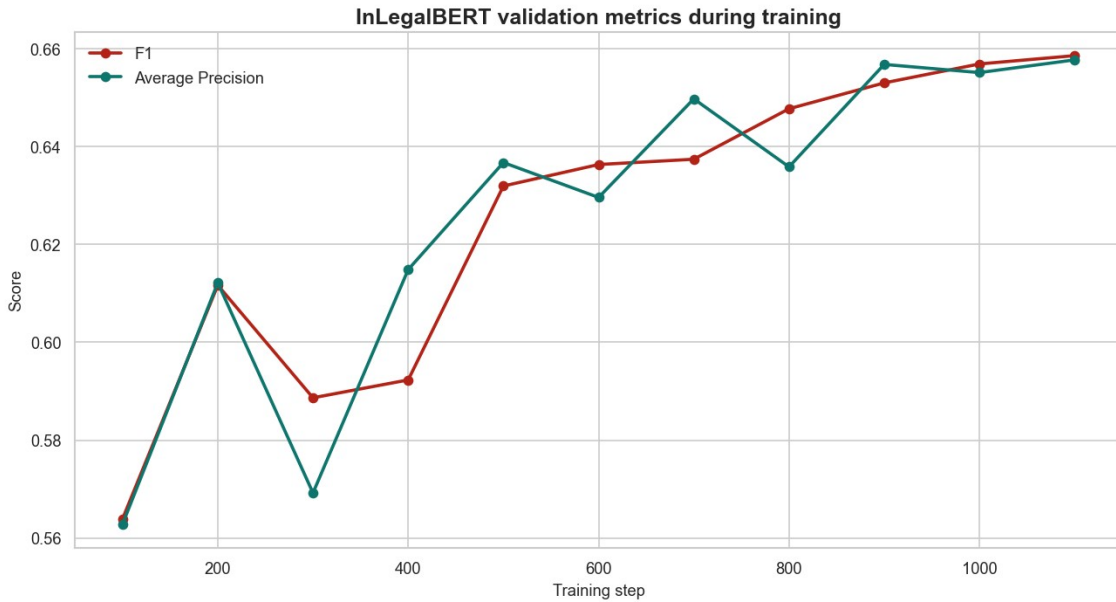


Figure 5.4: InLegalBERT validation F1 and Average Precision.

5.8 Pseudo-Label Evaluation

On Gemini-labelled evaluation, the reranker achieved MRR 0.8229, Recall@10 0.9732, MAP 0.7999 and nDCG@10 0.8200. Each value was substantially higher than the BM25 baseline on the same pseudo-label definition.

This comparison demonstrates task learning, but it is susceptible to label-source alignment: the model can learn the patterns Gemini used without learning the complete official relevance relation.

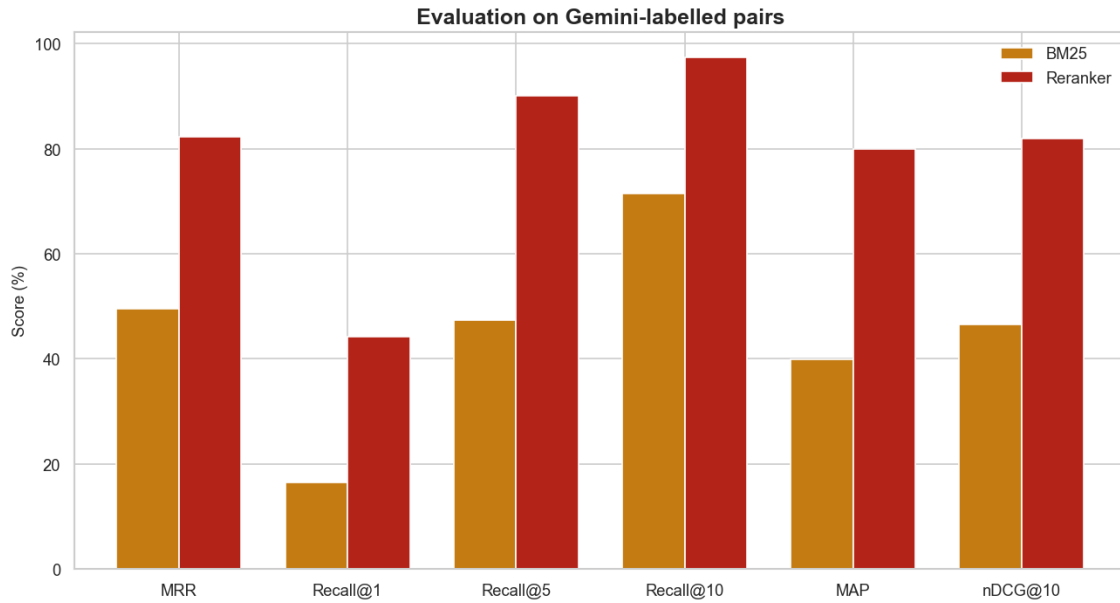


Figure 5.5: BM25 and reranker performance on Gemini-labelled pairs.

5.9 Official Test Result

The official comparison changed the conclusion. BM25 achieved Micro-F1@10 0.194777. The pure reranker achieved 0.177203 and therefore did not beat the lexical baseline.

The hybrid system achieved the best Micro-F1@10 of 0.209422. It retrieved 429 correct pairs from 1,727 official gold pairs, with Precision@10 0.181013 and Recall@10 0.248408.

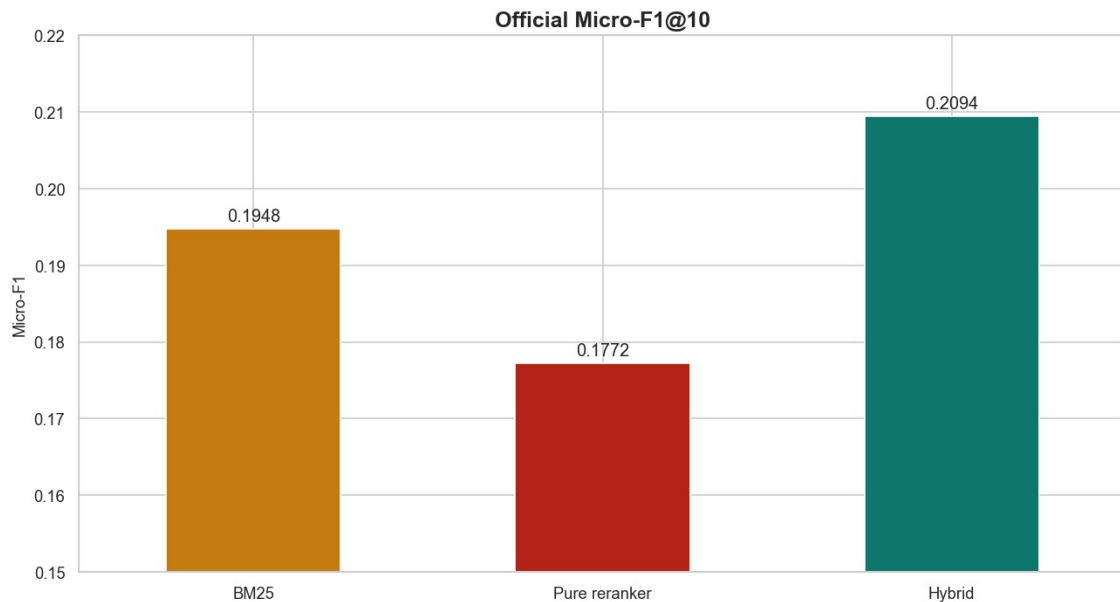


Figure 5.6: Official Micro-F1@10 for BM25, pure reranker and hybrid.

5.10 Hybrid Score

The best normalized fusion used 55% BM25 score and 45% reranker score. BM25 preserved exact statutory wording, citations and legal terminology. The reranker contributed semantic support between query Facts and candidate Ratio.

Fusion reduced the risk of relying entirely on either signal. The result also shows why pseudo-label evaluation must be separated from official evaluation.

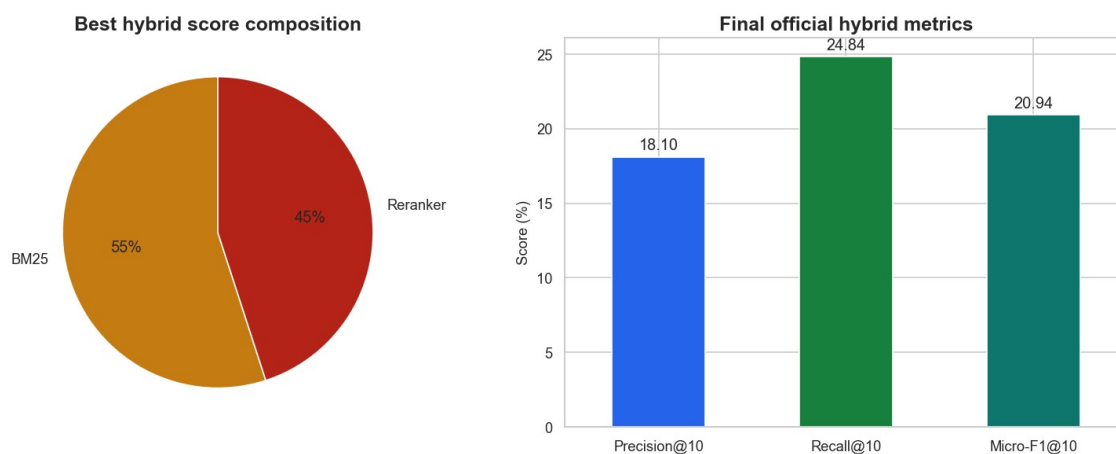


Figure 5.7: Best hybrid composition and official hybrid metrics.

5.11 Phase 2 Summary

Phase 2 successfully created a large entailment-style dataset and trained a cross-encoder that learned the generated task. The model's official performance, however, was weaker than BM25 when used alone.

The main Phase 2 outcome is the official hybrid result. Future work should train with more official positives, split data by query, preserve long role text through windows and compare BM25-based pools with the stronger Phase 1 TF-IDF baseline.

System	Official Micro-F1@10	Interpretation
BM25	0.194777	Strong exact-term baseline.
Pure reranker	0.177203	Did not transfer fully from pseudo-labels.
Hybrid	0.209422	Best official result.

5.12 Hierarchical Sliding-Window InLegalBERT

The standard cross-encoder truncates each pair to 512 tokens. The hierarchical notebook instead divided query Facts and candidate Ratios into overlapping windows, encoded several window pairs with law-ai/InLegalBERT and pooled their [CLS] representations through learned attention. The query-disjoint data split contained 33,177 training pairs and 5,897 validation pairs.

Measure	Saved result
Training schedule	One executed epoch in the supplied notebook
Validation MRR	0.3957
Official Precision@10	0.1705
Official Recall@10	0.2339
Official Micro-F1@10	0.1972
Official prediction coverage	237 queries

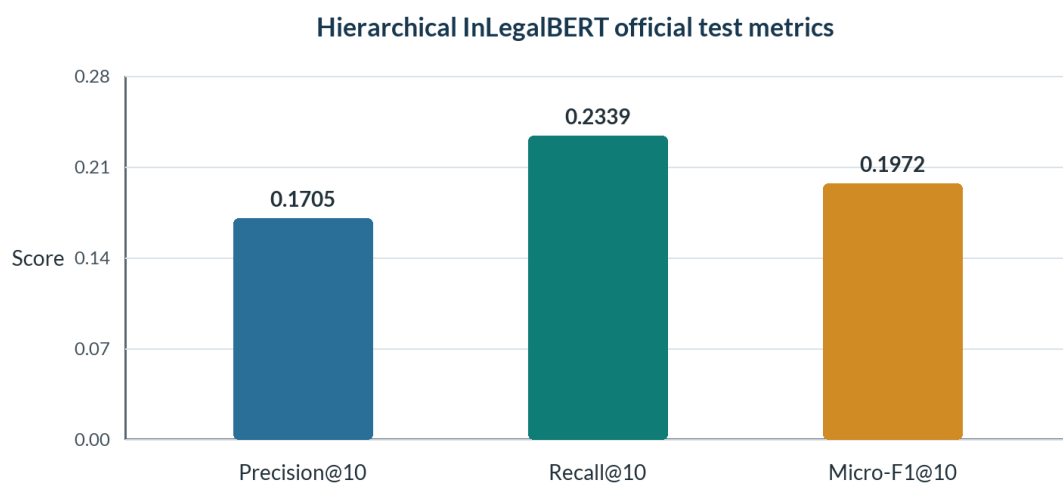


Figure 5.8: Official test metrics from the hierarchical InLegalBERT notebook.

The experiment confirms that windowed encoding is feasible, but the saved one-epoch run did not improve on the 55% BM25 and 45% reranker hybrid. More complete training and better window selection are therefore required before treating the hierarchical model as the final system.

CHAPTER 6 - PHASE 3: RHETORICAL-ROLE WEIGHTING AND LEARNING TO RANK

Phase 3 investigates which rhetorical-role matches support relevance and which create misleading similarity. We analysed separate positive and negative Gemini datasets, selected five role pairs, built a full sliding-window ModernBERT cache and trained a lightweight ranking head.

Gemini data was used for role-prior initialization and hard-negative selection. The actual positive relevance labels for training came from official train.json.

6.1 Meaning of Positive and Negative Importance

In the positive dataset, an importance score answers: how much did this role match contribute to genuine relevance? A score of five indicates decisive legal support.

In the negative dataset, a high importance score has the opposite interpretation: how strongly did this role create superficial or misleading overlap even though the official pair is not relevant?

Dataset	Rows	Interpretation
Positive	5,184	Role evidence contributing to relevant pairs.
Negative	5,181	Strong-looking overlap in non-relevant pairs.
Critical distinction: A high negative importance score is not evidence of relevance.		

6.2 Dataset Quality and Coverage

The executed results notebook checked row counts, unique queries, unique candidates, duplicate pairs, missing relevance scores and completeness of the ten importance columns.

Both datasets cover a similar number of unique pairs, but the negative dataset draws from a smaller candidate set because it focuses on hard negatives rather than arbitrary non-relevant combinations.

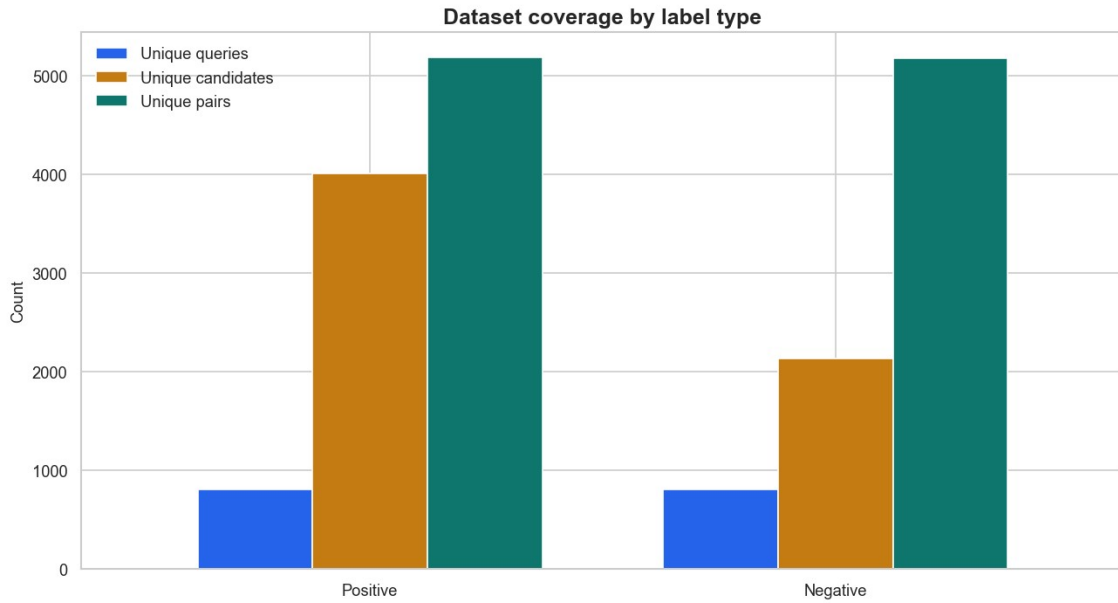


Figure 6.1: Positive and negative dataset coverage.

6.3 Positive Role Importance

Ratio-to-Ratio had the highest positive mean importance of 3.38. It appeared in 4,563 of 5,184 positive top-role lists, and 53.5% of positive pairs assigned it a score of at least four.

Fact-to-Fact and Precedent-to-Precedent followed. The result suggests that relevant prior cases combine legal reasoning, factual analogy and cited authority.

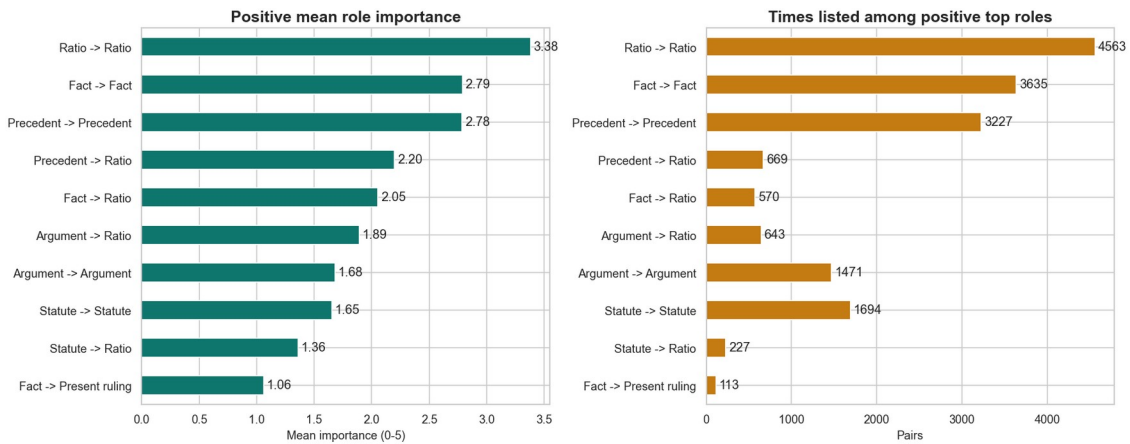


Figure 6.2: Positive mean importance and top-role frequency.

6.4 Negative Role Importance

Fact-to-Fact had the highest negative mean importance of 2.35 and was selected as the first misleading role 2,808 times. Statute-to-Statute was first 1,409 times.

These patterns show that similar events or shared statutory references can create strong lexical similarity without establishing the same controlling issue, procedural posture or ratio.

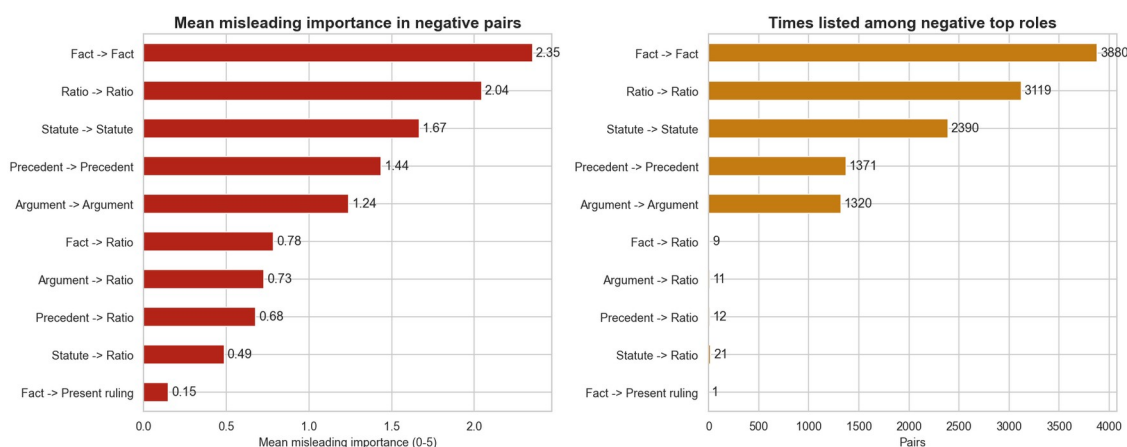


Figure 6.3: Mean misleading importance and negative top-role frequency.

6.5 Relevance-Score and Importance Distributions

The positive Gemini relevance scores had mean 77.32 and median 80. The negative scores had mean 27.84 and median 30. The distributions are well separated by construction, but the role-score distributions overlap.

Boxplots and violin plots show why averages are insufficient. Some negative pairs receive high Fact or Statute importance, while some positive pairs rely on a combination of moderate role signals.

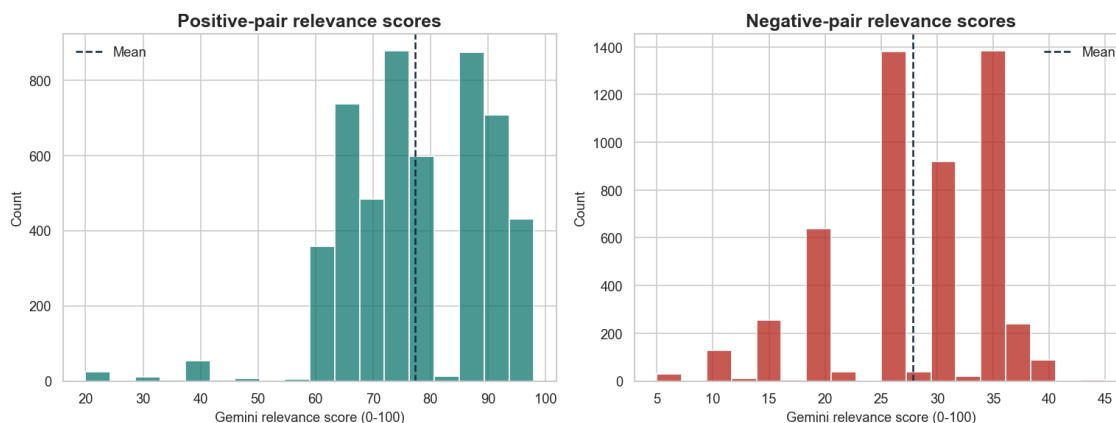


Figure 6.4: Positive and negative relevance-score distributions.

6.6 Role Co-occurrence and Correlation

Role co-occurrence heatmaps count how often two role pairs appear together in the same top-role explanation. Positive explanations commonly combine Ratio-to-Ratio, Fact-to-Fact and Precedent-to-Precedent.

Spearman correlations measure the monotonic relationship between ordinal role importance and Gemini relevance score. The analysis is descriptive and does not train a classifier.

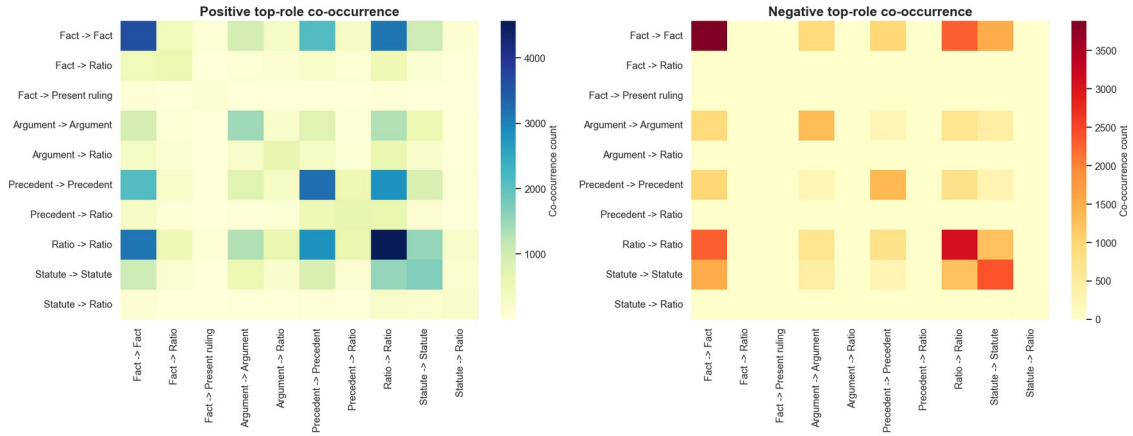


Figure 6.5: Positive and negative top-role co-occurrence heatmaps.

6.7 Positive-Negative Separation

We calculated mean differences, high-importance rate differences and Cohen's d for each role pair. Precedent-to-Ratio, Fact-to-Ratio and Ratio-to-Ratio showed strong positive-negative separation.

Effect sizes complement frequency counts: a role can be common in both datasets but still provide limited discrimination. Statute-to-Statute is an example of a frequent but potentially misleading signal.

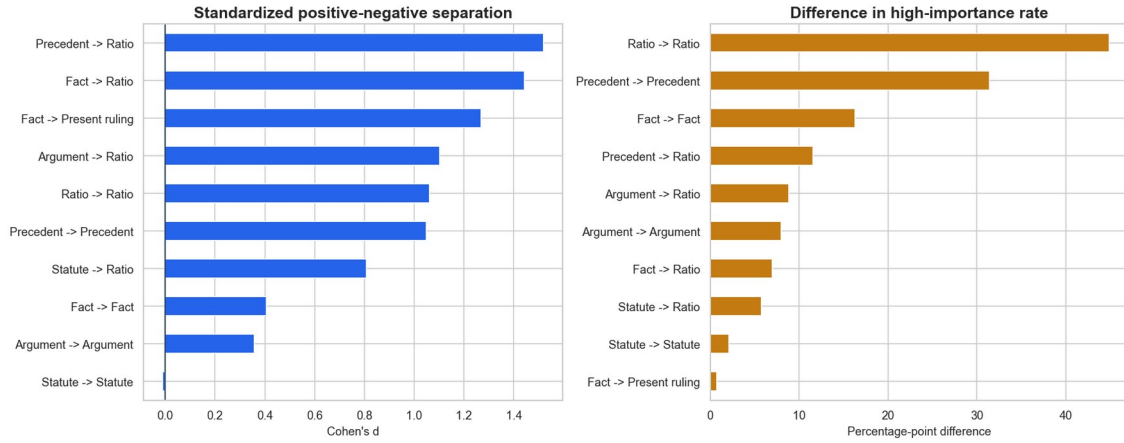


Figure 6.6: Standardized separation and high-importance-rate differences.

6.8 Selected Role Prior

Five role pairs were retained: Ratio-to-Ratio, Fact-to-Fact, Precedent-to-Precedent, Precedent-to-Ratio and Fact-to-Ratio. The initial normalized prior assigned 0.264564 to Ratio-to-Ratio and 0.250755 to Fact-to-Fact.

After training, the prior changed only slightly. Ratio-to-Ratio increased to 0.267124 and Fact-to-Fact increased to 0.252421. The model continued to emphasize the same three principal roles.

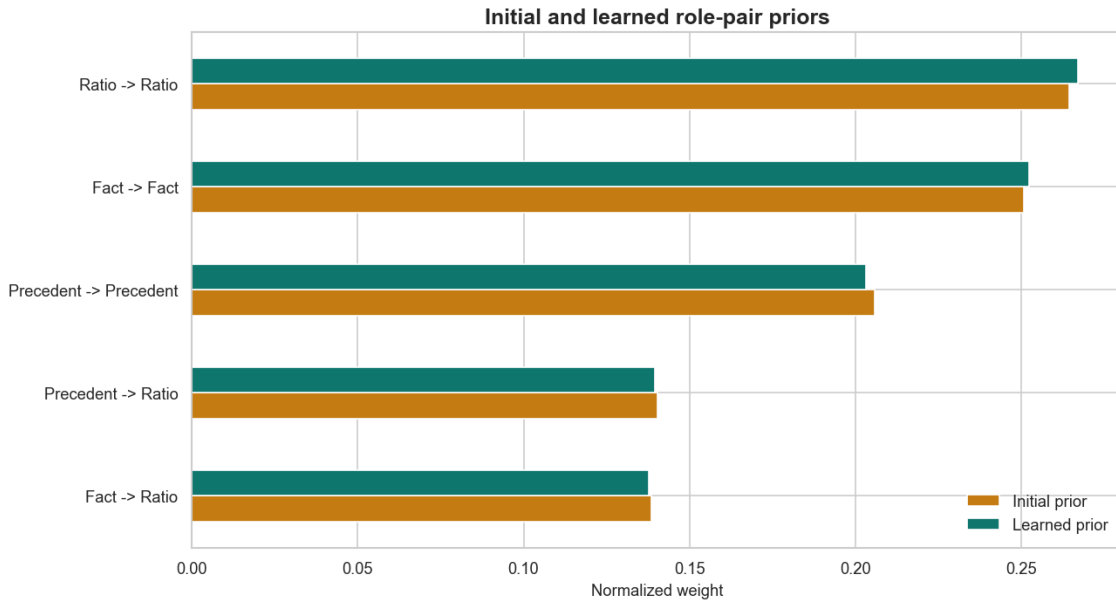


Figure 6.7: Initial and learned role-pair priors.

6.9 Official Positives and Query Groups

Official train.json contained 5,308 positive pairs across 817 queries. After checking that at least one active top-five role pair contained usable text, 5,179 positives across 806 queries and 4,006 positive candidates remained.

A query-level split assigned 669 queries to training and 137 to internal validation. Each group contained one positive and seven hard or random negatives, producing 35,112 training slots and 6,320 validation slots.

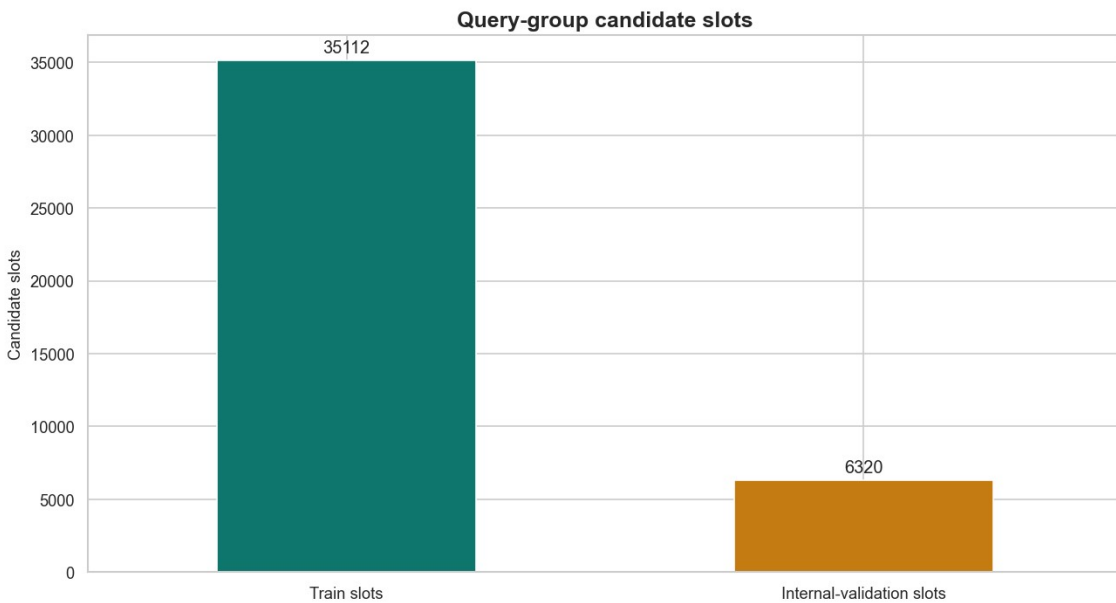


Figure 6.8: Candidate slots in the Phase 3 query groups.

6.10 Cached Full-Role-Text Architecture

Full selected role text can produce dozens or hundreds of 512-token windows for one query-candidate pair. End-to-end backpropagation through every window was not practical on a single NVIDIA L4 GPU with 22.04 GB VRAM.

We froze `answerdotai/ModernBERT-base`, encoded every sliding window once, stored the 768-dimensional CLS embeddings and trained only a lightweight role-aware ranking head. This preserved full selected role text while avoiding encoder activation memory during head training.

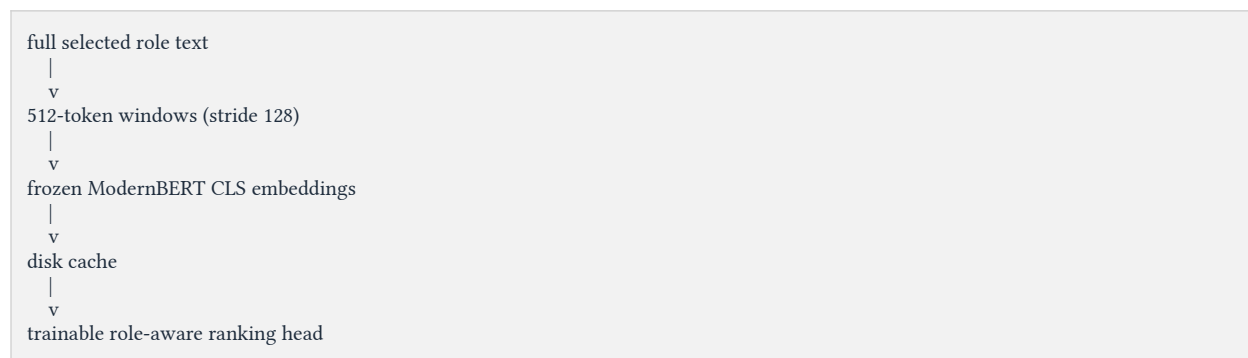


Figure 6.9: Full-role-text cached reranker architecture.

6.11 Cache Statistics

The cache contains 15,302 unique query-candidate pairs in 154 shards and occupies approximately 1,829.68 MB. The median pair produced 52 windows, the mean was 80.61, the 99th percentile was 480 and the maximum was 2,305.

Cache runtime depends on the number of windows, not only the number of pairs. Two 100-pair shards can therefore have substantially different encoding times.

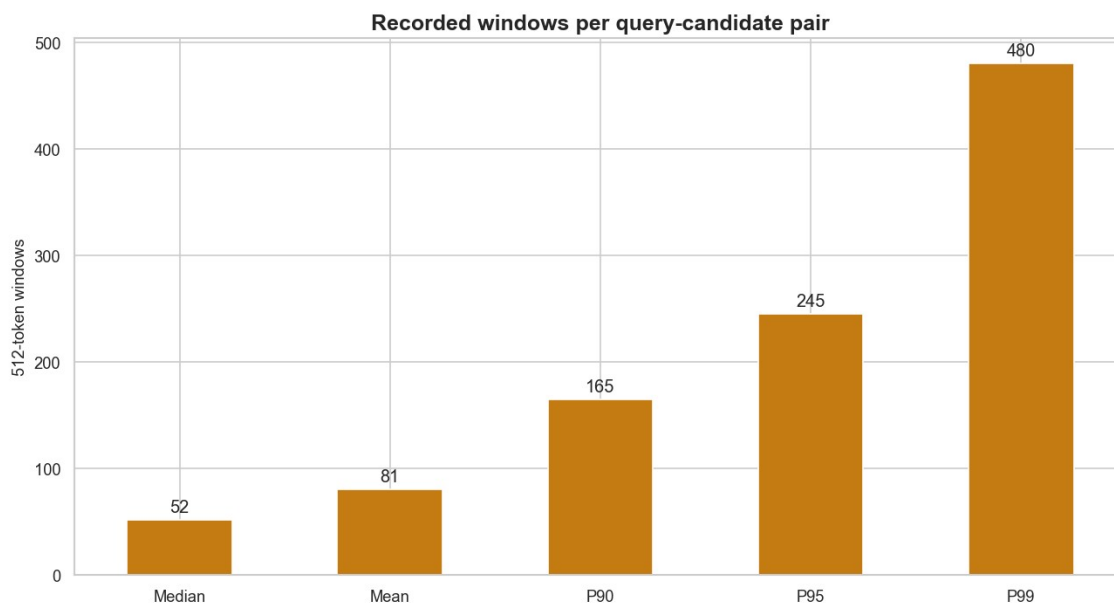


Figure 6.10: Recorded sliding-window distribution.

6.12 Ranking Head and Training Settings

The head contains a window scoring layer, dynamic gate, role-prior logits and a final bias. Only 298,760 parameters were trainable because ModernBERT remained frozen.

Setting	Value
Top windows per role aggregation	3
Maximum head epochs	6
Learning rate	5e-5
Weight decay	0.01
Dropout	0.05
Early-stopping patience	3

```

class CachedRankingHead(nn.Module):
    def __init__(self, hidden=768):
        self.window_score_head = nn.Linear(hidden, 1)
        self.dynamic_gate_head = nn.Linear(hidden, 1)
        self.role_prior_logits = nn.Parameter(prior.log())
        self.final_bias = nn.Parameter(torch.zeros(1))

```

Code 6.1: Simplified ranking-head structure.

6.13 Internal Training Results

The untrained internal baseline achieved MRR 0.311359, Hit@1 0.108861 and Hit@3 0.305063. Basic cached-head training improved MRR to 0.391349 at epoch two.

Query-level pairwise and listwise training improved MRR further to 0.401302, Hit@1 to 0.189873 and Hit@3 to 0.468354. LambdaRank did not improve beyond the starting query-pairwise checkpoint.

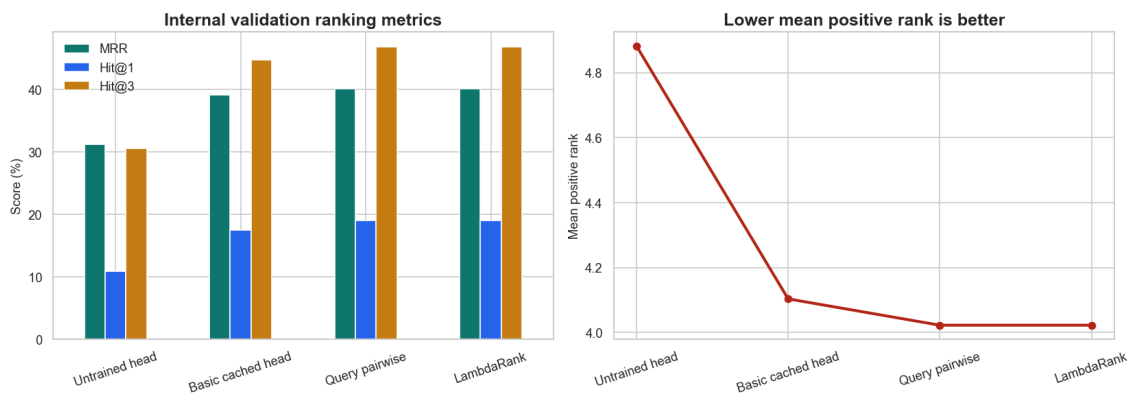


Figure 6.11: ModernBERT internal ranking progression.

6.14 Hard Negatives

The negative Gemini file contained 5,181 unique pairs across 807 queries and 2,134 candidates. After normalization, the hard-negative pool covered 806 queries, with a mean of 6.43 hard negatives per query and a maximum of 51.

BM25-ranked negatives are useful because they share lexical evidence with the query. Their Fact and Statute overlap teaches the ranker not to treat all strong surface similarity as legal relevance.

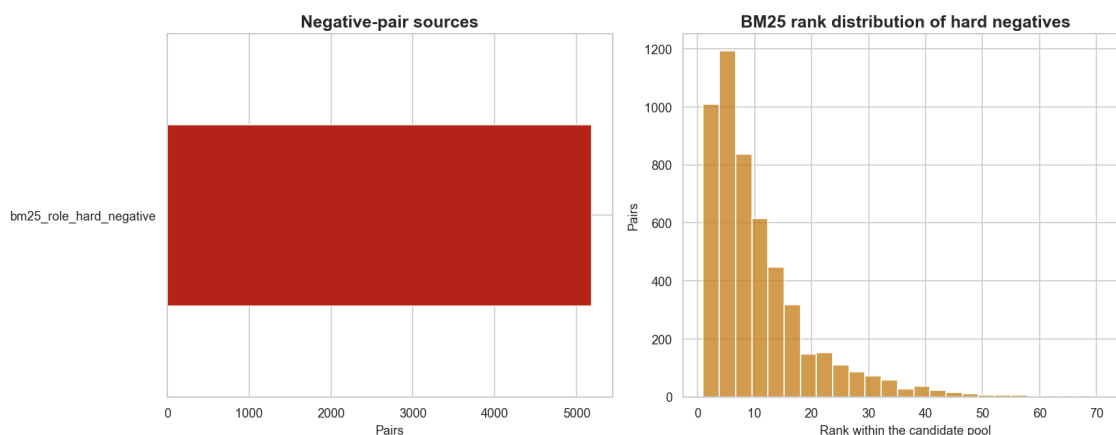


Figure 6.12: Negative sources and BM25 rank distribution.

6.15 Official Test Readiness

The official test contains 237 role-annotated query files and 1,727 candidate files. Brute-force scoring would require 409,299 pairs. A BM25 top-200 shortlist reduced this to 46,800 rows for 234 evaluated queries.

Of 1,727 gold pairs, 1,682 have usable text for at least one active top-five role pair, a usability rate of 97.39%. The remaining step is to create test-time CLS embeddings and score the shortlist with the saved query-pairwise checkpoint.

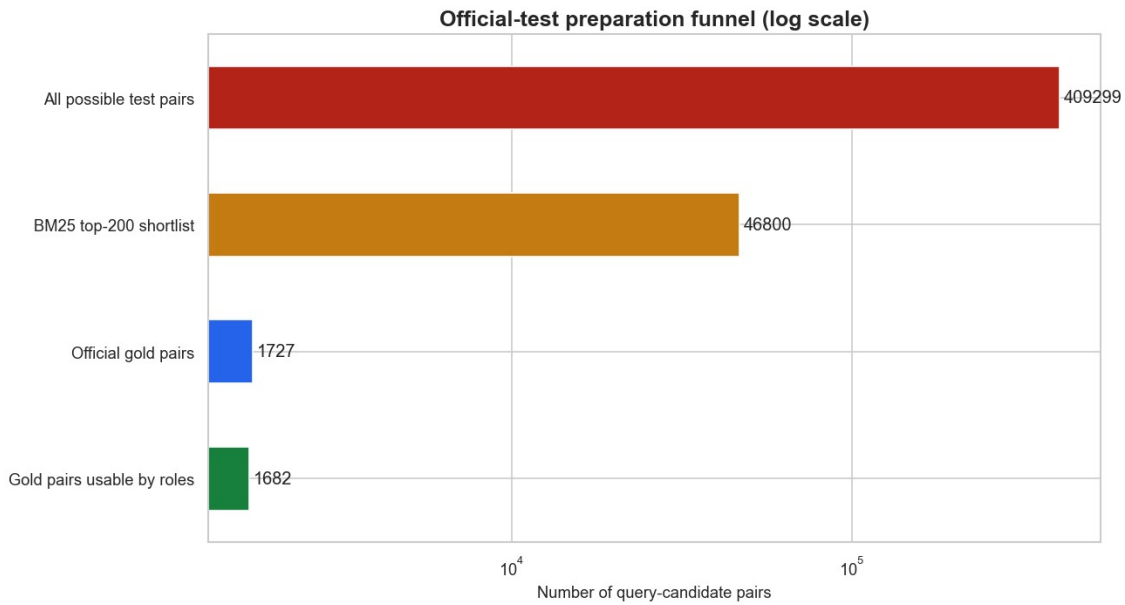


Figure 6.13: Official-test preparation funnel.

6.16 BM25 Official Baseline and Phase 3 Status

The Phase 3 summary recorded BM25-only official metrics at several cutoffs. At K=10, macro Recall was 0.301587, macro MRR was 0.480232 and Micro-F1 was 0.190804. These values are not neural-reranker results.

The provided evaluator contained a loop-unpacking bug and an incorrect printed K value. A corrected local version was prepared without modifying the original dataset file.

Official full-window neural reranking was postponed because the 46,800 test pairs require a new inference cache, approximately three times the size of the training pair cache.

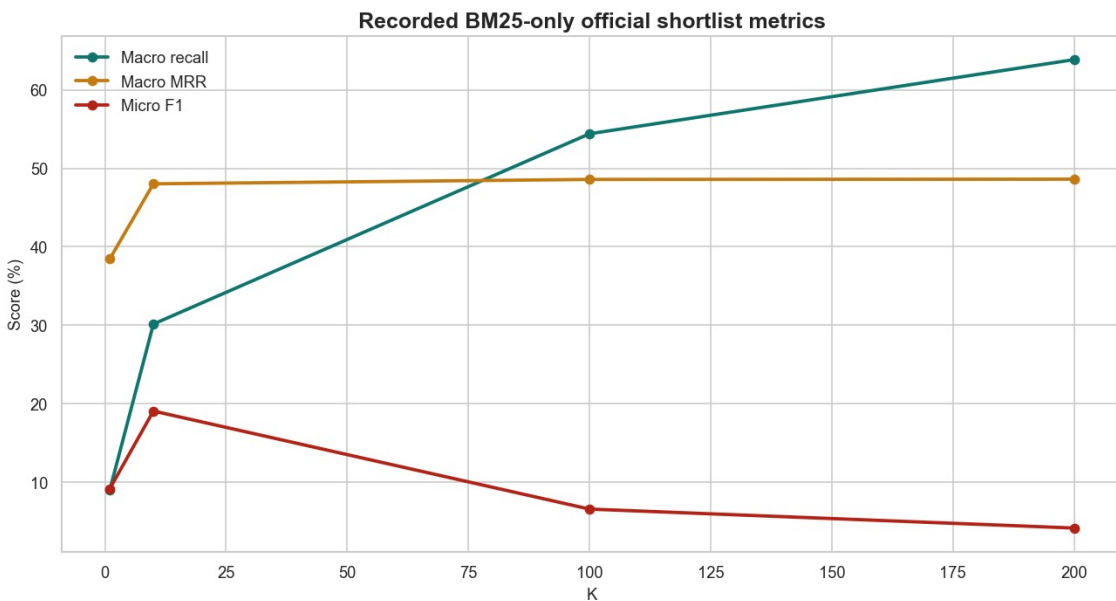


Figure 6.14: Recorded BM25-only official shortlist metrics.

6.17 Cached InLegalBERT LambdaRank Experiment

The final supplied notebook froze InLegalBERT role-pair embeddings and trained a compact 698,614-parameter reranking head with a LambdaRank-style, nDCG-weighted objective. It retained 807 valid query groups, with 686 used for training and 121 for validation. Query-grouped training compared candidates for the same query instead of treating every pair independently.

Evaluation setting	MRR	MAP / nDCG@10	Recall@10	Interpretation
Internal validation (full model)	0.9711	nDCG 0.9360	0.8825	Model-selection result; not official test performance
Official text-only test	0.1414	MAP 0.0615	0.4821	Across 237 test queries
70% neural + 30% BM25	0.1489	MAP 0.0654	0.4821	Small gain over neural-only ranking
Grid-search best: 100% BM25	0.4570	-	-	Strongest saved MRR in the weight sweep

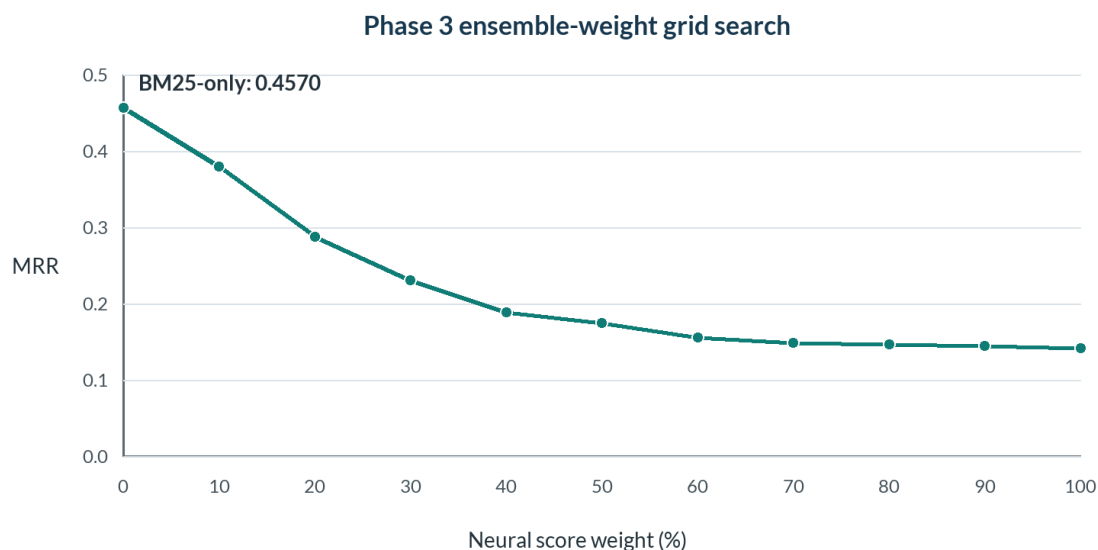


Figure 6.15: MRR obtained while varying the neural and BM25 ensemble weights.

The internal-validation score did not transfer to the official test collection. The neural head learned the cached training groups, but its ranking did not generalize to the official candidate pool. BM25 therefore remained the strongest saved Phase 3 test ranking, while the LambdaRank model was retained as an executed experimental branch.

6.18 Ten-Epoch Text-Only Training Check

A second branch in the same notebook trained the text-only LambdaRank model for ten epochs. The final two epochs are reproduced below as a compact run summary. Training loss continued to decrease, while validation MRR stayed at 0.9132 and the top-10 measures changed only slightly. These are query-grouped internal validation scores, not official test results.

Epoch	Average train loss	Optimizer steps	Validation MRR	nDCG@10	Recall@10
9	0.2944	43	0.9132	0.8330	0.8213
10	0.2819	43	0.9132	0.8328	0.8208

Table 6.4: Final epochs of the text-only LambdaRank training run.

CHAPTER 7 - IMPLEMENTATION AND REPRODUCIBILITY

The project was implemented as a collection of notebooks, datasets, saved CSV/JSON summaries and model artifacts. The notebooks reflect iterative experimentation rather than a single packaged application.

For reporting, we consolidated outputs into three result notebooks. These notebooks load saved metrics or data and produce tables and charts without loading models or rerunning retrieval, inference, training or testing.

7.1 End-to-End Workflow and Artifacts

The workflow begins with raw judgments and role annotations, proceeds through lexical retrieval and candidate pooling, and then applies entailment or role-aware models. Official evaluation remains the final authority.

- Raw and role-annotated legal text remains separate from generated labels.
- Official qrels are used for final claims.
- Large model caches are retained so training does not need to be repeated.
- Visualization notebooks contain no model-execution code.

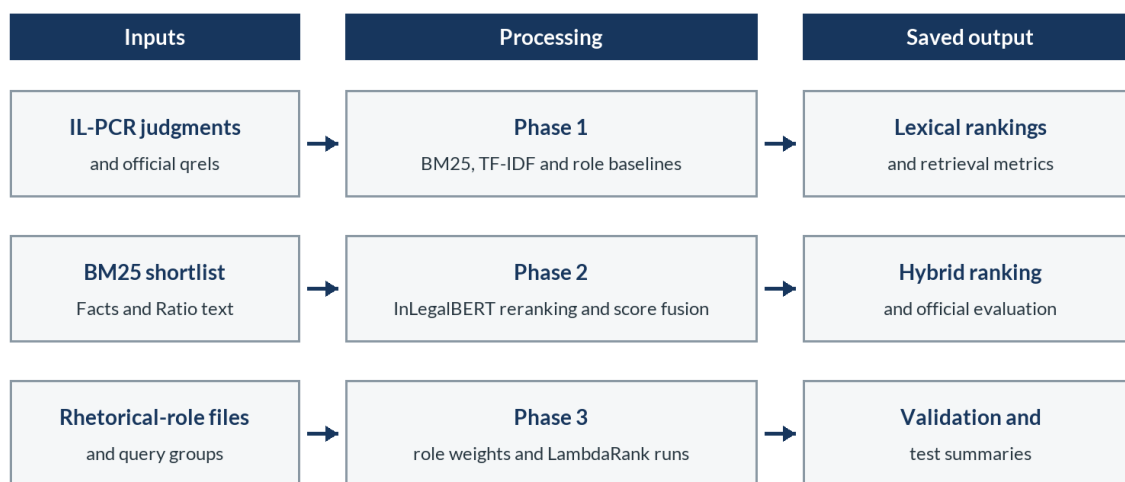


Figure 7.1: Consolidated project workflow.

7.2 Result Notebooks

The results folder contains phase1.ipynb, phase2.ipynb and phase3.ipynb. The notebooks were executed by the students and their saved plots are used throughout this report.

Notebook	Contents
phase1.ipynb	Dataset scale, lexical metrics, sample ranking, role coverage and learned weights.
phase2.ipynb	Dataset preparation, label imbalance, training history, pseudo and official results.

Notebook	Contents
phase3.ipynb	Role distributions, correlations, co-occurrence, effect sizes, priors, cache and readiness.
<pre>PROJECT_ROOT = find_project_root() positive = pd.read_csv(positive_path) negative = pd.read_csv(negative_path) # Descriptive aggregation only positive_stats = role_statistics(positive) negative_stats = role_statistics(negative)</pre>	

Code 7.1: Visualization-only result loading.

7.3 Ethical and Practical Considerations

The system is a research prototype for retrieval assistance. It should not be used to make legal decisions without professional review. A high similarity score does not establish that a precedent controls a case.

Generated labels can reproduce prompt bias and model bias. We therefore separate generated supervision from official relevance and report both pseudo-label and official results.

Legal judgments can contain sensitive personal information. Practical deployment would require access controls, data minimization, secure storage, audit logs and a clear retention policy.

- Preserve human review of retrieved cases and supporting passages.
- Expose why a case was retrieved through role-level evidence.
- Monitor failure cases involving shared statutes and superficial fact overlap.
- Do not claim an unfinished official neural evaluation as a completed result.

CHAPTER 8 - LIMITATIONS

The project achieved useful baselines and internal modelling results, but several parts remain incomplete or methodologically constrained.

Limitation	Effect
Phase 1 execution gap	TextRank, InLegalBERT dense retrieval and alpha sweeps lack saved outputs.
Retriever inconsistency	Later pools used BM25 although reproduced TF-IDF was stronger.
Pseudo-label mismatch	Gemini support labels do not exactly match official relevance.
Long-text truncation	Phase 2 cross-encoder may miss later reasoning passages.
Class imbalance	Positive entailment labels are a small minority.
Sampled validation	Phase 3 internal groups differ from full-corpus retrieval.
Pending official neural test	ModernBERT improvement is internal only.

Limitation	Effect
Computational cost	Full sliding-window inference requires a large test cache.

8.1 Future Enhancements

The next experiments should be prioritized according to how directly they resolve the current uncertainties.

- Execute every prepared Phase 1 method with the same qrels, cleaning and cutoffs.
- Compare BM25, TF-IDF and a lexical hybrid at identical shortlist sizes.
- Train entailment models with more official positives and query-group splits.
- Use windowed Facts-to-Ratio inputs rather than single 512-token truncation.
- Complete the Phase 3 test-time cache and official neural reranking without retraining.
- Evaluate score fusion between the Phase 2 hybrid and Phase 3 role-aware reranker.
- Add passage-level explanations showing the roles and windows responsible for each result.
- Package the pipeline with configuration files, deterministic seeds and automated evaluation.

Highest-priority continuation: Resume official Phase 3 test reranking from the saved query-pairwise checkpoint and BM25 top-200 shortlist.

CHAPTER 9 - CONCLUSION

This internship examined Indian legal prior-case retrieval through three connected perspectives: lexical matching, legal entailment and rhetorical-role weighting. The results show that exact legal wording remains essential, but legal structure and semantic support provide complementary evidence.

The reproduced Phase 1 experiment found TF-IDF to be the strongest full-text baseline under our settings. The role-aware experiments showed that Precedent-to-Precedent was the best isolated role, while Fact-to-Fact received the largest learned discriminative weight and a combined score improved retrieval.

Phase 2 demonstrated the difference between learning pseudo-labels and improving official retrieval. The pure InLegalBERT reranker performed strongly on Gemini-labelled evaluation but did not beat BM25 officially. The 55% BM25 and 45% reranker hybrid achieved the best official Micro-F1@10 of 0.209422.

Phase 3 provided the clearest interpretation of role evidence. Ratio agreement was central to positive relevance, whereas Facts and Statutes often created misleading similarity among negatives. ModernBERT improved the sampled internal ranking task, while the supplied InLegalBERT LambdaRank run exposed a large generalization gap: high internal validation did not translate into stronger official retrieval, and BM25 remained the most dependable saved test ranking.

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10. Project summaries and result files: phase2_results.json, phase2_training_history.csv, pcr_modernbert_l4_training_summary.pdf and results/phase1-3.ipynb.

APPENDIX: OUTPUTS AND CONTRIBUTION SUMMARY

A.1 Student Details

Student	Roll Number	Shared Contribution
Aniket Mani	205 124 013	Dataset analysis, experiments, evaluation, visualization and documentation.
Rishu Kumar	205 124 078	Dataset analysis, experiments, evaluation, visualization and documentation.

A.2 Final Status

The project completed reproducible lexical results, role-aware experiments, a Phase 2 official hybrid result, and executed hierarchical InLegalBERT and LambdaRank experiments. The ModernBERT full-window official run remains incomplete; the supplied InLegalBERT LambdaRank test was completed but did not outperform BM25.